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CASE STUDY : 2

Urban Air Quality in Indian Cities: EDA & Hypothesis Testing

P24CS904

DATA SCIENCE

THARUN MARAN R
Reg. No: 125262013
M.Tech CSE

SUBMITTED TO :
Dr. M. PRABHA
Associate Professor

1. Introduction

Air pollution is an important environmental and public-health challenge in many Indian cities. Fine particulate matter (PM2.5) and nitrogen dioxide (NO_2) are influenced by transport, industry, energy use, weather and other urban activities. Multi-year monitoring records can be used to describe pollution patterns, compare cities and seasons, and examine changes during major disruptions such as the 2020 COVID-19 lockdown.

This case study analyses CPCB-based city-day air-quality observations for 2015–2020. It applies exploratory data analysis (EDA), confidence intervals, hypothesis testing and pollutant correlation analysis. The purpose is statistical description and comparison, not proof that one event caused a change in air quality.

2. Problem Statement

Air-quality measurements vary across cities, dates and monitoring stations. Raw records may contain missing observations, duplicate records, inconsistent city or station names, and extreme values. A fair comparison of 2019 and 2020 therefore requires careful data preparation, consistent aggregation and a clearly defined comparison window. The project summarises PM2.5 and NO_2 by city and season, estimates uncertainty, compares the 2020 lockdown period with the corresponding 2019 dates, and examines relationships among pollutants.

3. Objectives

- Collect and prepare CPCB city-day air-quality observations for Indian cities from 2015–2020.
- Summarise PM2.5 and NO_2 distributions by city and season using descriptive statistics and plots.
- Calculate confidence intervals for pollutant summaries.
- Compare pollutant concentrations during a defined 2020 lockdown window with matching dates in 2019.
- Apply Welch's t-test or Mann–Whitney U test, as appropriate, and report effect sizes.
- Study pollutant correlations and explain why correlation does not establish causation.
- Present findings, limitations and implications in a five-minute briefing.

4. Dataset Description and Scope

The intended source is CPCB public air-quality monitoring data. The TUMI Data Portal provides a compiled India urban air-pollution dataset based on CPCB data with hourly and daily observations, including PM2.5, PM10, NO_2, NOx, SO_2, CO, O_3 and AQI. Confirm the exact resource, time coverage, units, cities and station coverage before analysis. The CPCB AQI portal and data repository are also official reference points.

Variable	Meaning / use
date	Observation date; used to derive year, month and season.
city / city_name	City identifier for city-wise comparisons.
station	Monitoring station identifier, if available.
PM2.5	Fine particulate matter concentration, commonly µg/m³.
NO_2	Nitrogen dioxide concentration, commonly µg/m³.
Other pollutants	PM10, SO_2, CO, O_3 and other measured variables, if present.

5. Research Questions and Hypotheses

Research question	Hypothesis
Did PM2.5 differ between the lockdown window in 2020 and the same dates in 2019?	H_0: The two periods have equal distributions/central tendencies. H_1: They differ.
Did NO_2 differ between the two periods?	H_0: The two periods have equal distributions/central tendencies. H_1: They differ.
Are pollutant concentrations associated?	H_0: Population correlation is zero. H_1: Population correlation is not zero.

Use a two-sided significance level of $\alpha = 0.05$ unless a different threshold is justified. Report effect sizes and confidence intervals with p-values. Statistical significance alone does not show whether a difference is environmentally important.

6. Methodology

6.1 Data Preparation and Quality Checks

- Load the downloaded CPCB-based file and inspect shape, columns, data types and date coverage.
- Convert date fields to datetime and pollutant concentrations to numeric values.
- Standardise city, station and pollutant names by trimming whitespace and using consistent labels.
- Identify missing values and source-specific missing codes; convert them to NaN.
- Check duplicates using available station, date and pollutant identifiers; investigate before removal.
- Inspect negative, impossible or unusually high values. Do not automatically delete plausible high pollution readings.
- Filter to 2015–2020 and record the cities and stations retained.

6.2 Season Classification

For a reproducible grouping, use Winter (December–February), Summer/Pre-monsoon (March–May), Monsoon (June–September) and Post-monsoon (October–November). This is an analytical convention; seasonal patterns vary by region, so state the definition clearly.

Season	Months
Winter	December, January, February
Summer / Pre-monsoon	March, April, May
Monsoon	June, July, August, September
Post-monsoon	October, November

6.3 Exploratory Data Analysis

Calculate count, mean, median, standard deviation, minimum, maximum and quartiles for PM2.5 and NO_2. Group results by city and season. Recommended visualisations include city-wise boxplots, seasonal comparison plots, histograms and daily time-series plots. Always show sample counts because cities may have different numbers of valid observations.

6.4 Confidence Intervals

A 95% confidence interval for a mean can be calculated using the t distribution: sample mean ± t critical value × (sample standard deviation / square root of n). It describes uncertainty in the estimated mean; it does not mean that 95% of individual daily measurements fall within the interval. For skewed data, consider bootstrap confidence intervals and specify whether the interval estimates a mean or median.

6.5 2019 versus 2020 Lockdown Comparison

Define the comparison window before looking at the results. A useful national reference period is 25 March–31 May 2020, compared with 25 March–31 May 2019. The Government of India announced that the nationwide order would take effect from 25 March 2020. Use the same cities and, where possible, the same monitoring stations in both periods. If a narrower lockdown phase is selected, justify and document it.

Welch's independent-samples t-test compares means without assuming equal variances. Mann–Whitney U is a rank-based alternative for skewed or outlier-sensitive independent samples; it is not strictly a test of medians unless distribution shapes are similar. Daily observations may be serially correlated, and stations within a city may not be independent. Interpret simple tests cautiously; stronger options include matched city/station comparisons, block bootstrap or regression with time and location controls.

6.6 Correlation Analysis

Use Pearson correlation to assess linear association when appropriate, and Spearman rank correlation for monotonic relationships or data sensitive to skewness and outliers. A correlation matrix or heatmap can summarise pollutant relationships. Correlation does not prove causation: pollutants may share sources, while weather, season, wind, rainfall, traffic and industrial activity can influence both.

7. Statistical Test Decision Guide

Method	Use when	Report
Welch t-test	Comparing means of two independent groups; unequal variances allowed.	Mean difference, 95% CI, t statistic, degrees of freedom, p-value.
Mann–Whitney U	Comparing independent groups with skewed/outlier-sensitive data.	U statistic, p-value, effect size; explain rank-based interpretation.
Pearson correlation	Assessing linear association between numeric pollutants.	r, p-value and sample size.
Spearman correlation	Assessing monotonic association or rank-based relationship.	ρ, p-value and sample size.

8. Expected Outputs

Deliverable	Contents
EDA notebook	Data import, cleaning, quality checks, city/season summaries and plots.
Statistical test summary	Period definitions, sample sizes, confidence intervals, test statistics, p-values and effect sizes.
Correlation analysis	Pollutant correlation matrix/heatmap and interpretation.
Five-minute briefing	Research question, data, methods, key findings, limitations and conclusion.
Reproducibility record	Dataset source, download date, filters, missing-data decisions and

9. Interpretation Framework

After executing the notebook, interpret results in this order: describe city and seasonal patterns; compare means and medians; inspect confidence interval widths and sample sizes; report whether the period comparison is statistically significant; state the direction and magnitude of any difference; and discuss confounding factors. Do not insert numerical findings until they have been calculated from the actual dataset.

Result pattern	Careful interpretation
p-value < 0.05	Evidence against the null under the test assumptions; report effect size and limitations.
p-value ≥ 0.05	Insufficient evidence to reject the null; this does not prove the periods are identical.
Positive correlation	Pollutants tend to increase together in the sample; shared sources may explain this.
Negative correlation	Higher values of one tend to accompany lower values of the other; investigate weather and sampling.

10. Limitations

- Monitoring coverage and data completeness may differ across cities and years.
- City averages can hide variation between stations and neighbourhoods.
- Daily measurements may be temporally autocorrelated, violating simple independent-sample assumptions.
- Weather and seasonal conditions differ between years and can influence pollutant concentrations.
- Changes in monitoring equipment, station availability or reporting practices may affect comparability.
- The lockdown coincided with many other changes; a simple before/after comparison cannot attribute a difference to lockdown alone.
- Findings from monitored locations should not automatically be generalised to every Indian city or resident.

11. SDG Relevance

SDG 11 (Sustainable Cities and Communities) is relevant because air-quality monitoring supports healthier and more sustainable urban environments. SDG 13 (Climate Action) is relevant to environmental awareness and policy discussion, while recognising that local air pollutants and greenhouse gases are different categories and should not be conflated. This project contributes descriptive evidence rather than directly measuring climate mitigation.

12. Conclusion

This case study presents a reproducible workflow for analysing CPCB-based city-day air-quality observations from 2015 to 2020. EDA describes PM2.5 and NO_2 across cities and seasons; confidence intervals quantify uncertainty; hypothesis tests compare a pre-defined 2020 lockdown window with matching dates in 2019; and correlation analysis examines pollutant relationships. Reliable interpretation requires attention to missing data, sample size, temporal dependence, meteorology and monitoring coverage. Most importantly, statistical association is not proof of causation.

13. References and Data Sources

1. Central Pollution Control Board (CPCB), National Air Quality Index and public data repository: https://airquality.cpcb.gov.in/AQI_India/
2. Transformative Urban Mobility Initiative (TUMI), Urban air pollution measurements, India, 2015–2020 (compiled from CPCB data): https://portal.transport-data.org/@tumi/urban-air-pollution-india-2015-2020
3. Government of India, Ministry of Home Affairs / Press Information Bureau. Government orders prescribing lockdown, 24 March 2020: https://www.pib.gov.in/PressReleasePage.aspx?PRID=1607997
4. World Health Organization. WHO Global Air Quality Guidelines (2021): https://www.who.int/publications/i/item/9789240034228
5. SciPy documentation, statistical tests: https://docs.scipy.org/doc/scipy/reference/stats.html
6. Statsmodels documentation: https://www.statsmodels.org/
7. Pandas documentation: https://pandas.pydata.org/docs/